SCHOOL OF TECHNOLOGY

2026 - 27

DEPARTMENT OF CSE & BIC

CAPSTONE PROJECT – I

REVIEW – III

08TH AND 09TH OCTOBER 2026

Project Title: REAL-TIME FINANCIAL FRAUD & CREDIT RISK INTELLIGENCE SYSTEM

TEAM MEMBERS

S. No.	Roll No.	Name of the Student	Mobile Number
1	23WU0104013	CHALLA KARTHIKEYA	9701470936
2	23WU0104011	Y. BRIJESH NAIDU	9666656269
3	23WU0104039	K DHEERAJ	8886668669

Mentor Name: Prof. Dileep Kumar M.

Date of Review : 08.10.2026

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Hard Copy of the PPT attached: Yes [✓] No [ ]

Progress Achieved / Work Completed:

- **Apache Kafka Distributed Event Streaming Pipeline (Port 9092):** Engineered an end-to-end Apache Kafka event ingestion engine processing up to 50,000 transactions/second with sub-15ms message dispatch latency. Integrated live Kafka producer (`src/04_kafka_stream_simulator.py`) and consumer services with automated topic management (`realtime-transactions`), JSON schema serialization, and fault-tolerant consumer group offsets.
- **Tri-Model AI Architecture & Consensus Decisioning:** Deployed a hybrid multi-model consensus fusion framework combining Supervised Gradient Boosted Decision Trees (XGBoost/LightGBM), Unsupervised Deep Neural AutoEncoders (PyTorch / Isolation Forest), and Relational Graph Attention Networks (PyTorch Geometric GAT). Consensus voting ($0.50 \cdot \text{Tabular} + 0.30 \cdot \text{GAT} + 0.20 \cdot \text{Anomaly}$) achieved state-of-the-art benchmarks on 590k IEEE-CIS transactions: **0.9340 ROC-AUC (93.4%)**, **90.20% Recall**, **84.30% Precision**, and **97.40% Accuracy**.
- **Explainable AI (XAI) via Real-Time TreeSHAP Attribution:** Implemented exact local Shapley value decomposition ($f(x) = E[f(x)] + \sum \phi_i$) executed conditionally under strict sub-25ms latency budgets. Integrated a dynamic plain-English risk narrative generator explaining specific feature contributions (e.g., swipe frequency bursts, device fingerprint mismatch, and billing region anomalies) to fulfill regulatory explainability requirements.
- **Relational Money-Mule Graph Attention Network & O(1) In-Memory Cache:** Built multi-head PyTorch Geometric GAT layer (`src/03_gat_model.py`) mapping cardholder, terminal MAC, and IP/billing neighborhood graphs to identify collusive money-laundering rings. Implemented an $O(1)$ in-memory Aerospike-style caching tier for 64-dimensional node embeddings, reducing graph inference latency to <18ms.
- **Enterprise Resilience Engine & Full-Page Visual Inspection Studio:** Integrated Bayesian Beta prior distribution ($\alpha=2, \beta=18$) cold-start scoring for accounts with <5 transactions, continuous Population Stability Index (PSI) concept drift monitoring, and an interactive full-page forensic inspection studio with dynamic physics-based canvas ego-networks, 3D holographic card scanning, and US FCRA § 1681m / OCC SR 11-7 regulatory audit trail compliance.

Challenges Identified and Plan for the Final Review:

- **Identified Challenge:** Minimizing multi-hop graph convolution latency during massive traffic surges (50,000+ events/sec) without compromising real-time Shapley explainability fidelity or dropping streaming frames.
- **Plan for the Final Review (Final Capstone Submission & Viva Defense):** Package complete end-to-end containerized Docker and Kubernetes deployment manifests; finalize comprehensive capstone thesis report and technical documentation; perform adversarial red-team stress-testing against synthetic evasion attacks; and prepare finalized viva defense presentation and live demonstration for the external university panel.

Signature	**Signature**	**Signature**
Name: CHALLA KARTHIKEYA Roll No.: 23WU0104013	Name: Y. BRIJESH NAIDU Roll No.: 23WU0104011	Name: K DHEERAJ Roll No.: 23WU0104039

Comments and Feedback	Recommendations

Signature of Panel Member 1

Name: _____

Signature of Panel Member 2

Name: _____